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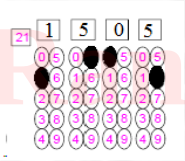
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 22-03-2026**

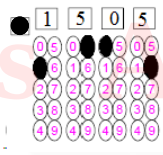
Time: 09.00AM TO 12.00PM **GTM-26** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below





Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. If $A=\begin{bmatrix}3 & 1\\ 7 & 5\end{bmatrix}$ and $A^2+xI=yA$ then twice the trace of the matrix $\begin{pmatrix}y & 2\\ 1 & x\end{pmatrix}$ is

1) 22 **2)** 16 **3)** 32 **4)** 24

2. If the locus of a point that divides a chord of slope 2 of the parabola $y^2=4x$ internally in the ratio 1:2 is a parabola and the vertex of this parabola is (α,β) then $18(\alpha+\beta)=$

1) 20 **2)** 10 **3)** 8 **4)** 16

3. If the area bounded by the curves $y=x^2, y=-x^2, y^2=4x-3$ is $\frac{a}{b}$,where H.C.F $(a,b)=1$, then $b^2-a^2=$

1) 15 **2)** 8 **3)** 10 **4)** 3

4. If $I_1=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$, $I_2=\int_0^{\pi}x\sin^4 x\,dx$ then $I_1:2I_2=$

1) 1:2 **2)** 2:3 **3)** 2:1 **4)** 3:2

5. Consider the regular polygon, which has 20 diagonals then match the following Column

	Column- I		Column –II
A)	Number of triangles joining the vertices of the polygon	**I)**	32
B)	Number of points of intersection of diagonal which lies in side the polygon	**II)**	70
C)	Number of triangles in which exactly one side is common with the given polygon	**III)**	8
D)	Number of triangles in which exactly two sides are common with the given polygon	**IV)**	56

1) A-II,B-III,C-I,D-IV **2)** A-IV,B-I,C-III,D-II

3) A-IV,B-III,C-I,D-II **4)** A-IV,B-II,C-I,D-III

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6. If the points $(2,3),(0,2),(4,5)$ and $(0,K)$ are concyclic then sum of possible values of K is

1) 21 **2)** 19 **3)** 15 **4)** 13

7. If the domain of the function $f(x)=\sqrt{\frac{4-x^2}{2+[x]}}$ is $(-\infty,\alpha)\cup[\beta,\gamma]$ then $\alpha+\beta+\gamma=$

1) 0 **2)** 1 **3)** -2 **4)** -1

8. If $P(m+n,2)=90$ and $P(m-n,2)=30$ and $\lambda=m^2-n^2$, then the number of positive divisors of λ is

1) 24 **2)** 15 **3)** 12 **4)** 18

9. In ΔABC, if one vertex $A(4,-1)$ and $x-1=0$ and $x-y-1=0$ are the bisectors of angles B and C respectively and the vertex $B(\alpha,\beta)$ then the quadratic equation having roots α,β is _____

1) $x^2-7x+10=0$ **2)** $x^2-6x+5=0$ **3)** $x^2-5x+6=0$ **4)** $6x^2-5x+1=0$

10. Let PQR be a triangle. Let $\bar{a}=\overline{QR},\bar{b}=\overline{RP},\bar{c}=\overline{PQ},|\bar{a}|=12,|\bar{b}|=43$ and $\bar{b}.\bar{c}=24$, then

A) $\frac{|\bar{c}|^2}{2}-|\bar{a}|=12$ B) $\frac{|\bar{c}|^2}{2}+|\bar{a}|=30$ C) $|\bar{a}\times\bar{b}+\bar{c}\times\bar{a}|=24\sqrt{3}$ D) $\bar{a}.\bar{b}=-72$

Choose the correct answer from the options given below

1) A,C and D only **2)** B and C only **3)** A and D only **4)** C only

11. If $|z-2+i|\le 2$, where $i=\sqrt{-1}$ then the sum of the greatest and least value of $|z|$ is

1) $\frac{\sqrt{5}}{2}$ **2)** $4\sqrt{5}$ **3)** $2\sqrt{5}$ **4)** $\sqrt{5}$

12. The number of values $\theta\in\left[\frac{-\pi}{2},\frac{4\pi}{3}\right]$ which satisfies the system of equations.

$2\sin^2\theta+\sin^2 2\theta=2$ and $\sin 2\theta+\cos 2\theta=\tan\theta$ is

1) 4 **2)** 3 **3)** 2 **4)** 1

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13. If $\int \frac{dx}{\sqrt{x}+\sqrt[3]{x}} = a\left(\sqrt{x}\right)+b\left(\sqrt[3]{x}\right)+c\left(\sqrt[6]{x}\right)+d\log\left(\sqrt[6]{x}+1\right)$+e, e being an arbitrary constant, then the value of $2a+3b+4c-5d=$

1) 0 **2)** 67 **3)** 49 **4)** 58

14. Statement: I :- If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$ then $4x^2-4xy\cos\alpha+y^2=2\sin^2\alpha$

Statement: II :- $\sum_{r=0}^{\infty}\tan^{-1}\left(\frac{1}{1+r+r^2}\right)=\frac{\pi}{4}$

1) Both statement- I and statement-II are true
2) Statement- I is true and statement-II is false
3) Statement –I is false and statement-II is true
4) Both statement- I and statement –II are false

15. If the value of $\lim_{x\to 0}\frac{1}{x^{18}}\left(1-\cos\left(\frac{x^3}{3}\right)-\cos\left(\frac{x^6}{6}\right)+\cos\left(\frac{x^3}{3}\right)\cos\left(\frac{x^6}{6}\right)\right)$

is λ^2 then $108\lambda=$____ (when $\lambda>0$)

1) 36 **2)** 3 **3)** 6 **4)** 2

16. Let the sum $\sum_{n=1}^{10}\frac{1}{n(n+1)(n+2)}$ written in the rational term be $\frac{p}{q}$ (where p and q are coprime) then the value of $q+p=$

1) 64 **2)** 130 **3)** 318 **4)**329

17. The number of non empty equivalence relations on the set $\{1,2,3,4\}$ is

1) 16 **2)** 15 **3)** 2^{12} **4)** 2^5

18. The mean and S.D of 15 observations ware found to be 12 and 3 respectively on rechecking it was found that an observation was read as 10 in place of 12.

If μ and σ^2 denote the mean and variance of the correct observations respectively then $30\left(\mu^2+\sigma^2\right)=$

1) 4678 **2)** 4576 **3)** 5478 **4)** 5678

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19. The thrice the length of the chord of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ whose mid point is $\left(\sqrt{2},\frac{4}{3}\right)$ is____

1) $\frac{2\sqrt{22}}{3}$ **2)** $3\sqrt{22}$ **3)** $2\sqrt{22}$ **4)** $6\sqrt{22}$

20. If $\frac{dy}{dx}+\frac{3^{x-y}\left(3^y-1\right)}{3^x-1}=0, x, y>0, y(1)=1$ then $y(2)=$ ___

1) $2-\log_3^2$ **2)** $1-\log_3^2$ **3)** $1-\log_3^4$ **4)** $2-\log_3^5$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Three critics review a book odds in favour of the book are 5:2,4:3 and 3:4 respectively for the three critics. If the probability that majority are in favour of the book is $\frac{11a}{343}$ then $a=$ _____

22. The number of irrational terms in the expansion of $\left(5^{1/2}+7^{1/8}\right)^{1029}+\left(5^{1/2}-7^{1/8}\right)^{1029}$ is

23. If the shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$ is λ, then $\lambda^2=$ ___

24. Consider the function $f(x)=\max\{|\sin x|,|\cos x|\}\ \forall x\in[0,3\pi]$, if λ is the number of points at which $f(x)$ is non- differentiable then $5\lambda^2=$ ___

25. For $\alpha,\beta\in R$, suppose the system of linear equations $x-y+z=5, 2x+2y+\lambda_1 z=8$, $3x-y+4z=\lambda_2$, has infinitely many solutions then distance of the point (λ_1,λ_2) from the line $3x-4y+4=0$ is _____

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. **Assertion:** The force with which two charges attract or repel each other are not affected by the presence of a third charge

Reason: Force on any charge due to number of other charges is the vector sum of all the forces on that charge due to other charges, taken one at a time

1) If both assertion and reason are true and reason is the correct explanation of assertion

2) If both assertion and reason are true but reason is not the correct explanation of assertion

3) If assertion is true but reason is false

4) If both assertion and reason are true

27. **Statement –I: -** The force on a charged particle moving in a uniform magnetic field is independent of the velocity of the particle.

Statement –II:- The force on the particle is given by $\bar{F}=q\left(\bar{v}\times\bar{B}\right)$

1) Both statement- I and statement-II are true

2) Statement- I is true and statement-II is false

3) Statement –I is false and statement-II is true

4) Both statement- I and statement –II are false

28. A concavo - convex lens of refractive index 1.5 and the radii of curvature of its surfaces are 30cm and 20cm respectively. The concave surface is upwards and is filled with a liquid of refractive index 1.3. The focal length of the liquid- glass combination will be.

1) $\frac{500}{11}cm$ **2)** $\frac{800}{11}cm$ **3)** $\frac{700}{11}cm$ **4)** $\frac{600}{11}cm$

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29. The interference pattern is obtained with two coherent light sources of intensity ratio 4:1 and the ratio $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is $\frac{5}{x}$. Then value of x will be equal to

1) 3 **2)** 4 **3)** 2 **4)** 1

30. **Statement –I:-** In a meter bridge experiment, null point for an unknown resistance is measured. Now the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance.

Statement –II:- Resistance of a metal increases with increase in temperature.

1) Statement – I is true, statement –II is true, statement II is a correct explanation for statement –I

2) Statement – I is true, statement –II is true, statement II is not a correct explanation for statement –I

3) Statement- I is true statement-II is false

4) Statement- I is false statement-II is true

31. Pressure depends on distance as, $P=\frac{\alpha}{\beta}\exp\left(-\frac{\alpha z}{K\theta}\right)$. Where α,β are constants, z is distance, K is Boltzman's constant and θ is temperature. The dimension of β are

1) $M^0L^0T^0$ **2)** $M^{-1}L^{-1}T^{-1}$ **3)** $M^0L^2T^0$ **4)** $M^{-1}L^1T^2$

32. Given below are two statements one is labeled as Assertion (A) and the other is labeled as Reason (R)

Assertion (A):- The binding energy per nucleon is found to be practically independent of the atomic number A, for nuclei with mass number between 30 and 170

Reason (R):- Nuclear force is long range

In this light of the above statements, choose the correct answer from the option given below,

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1) (A) is false but (R) is true

2) (A) is true but (R) is false

3) Both (A) and (R) are true and (R) is the correct explanation of (A)

4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

33. The position of a particle as a function of time 't' is given by $x(t) = at + bt^2 - ct^3$ where a, b, c are constants. When the particle attains zero acceleration then its velocity will be

1) $a + \frac{b^2}{4c}$ **2)** $a + \frac{b^2}{3c}$ **3)** $a + \frac{b^2}{c}$ **4)** $a + \frac{b^2}{2c}$

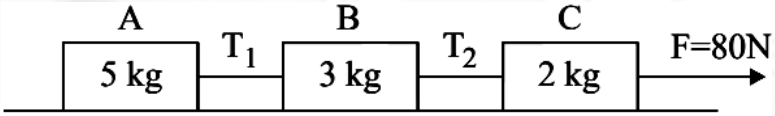
34. In a semiconductor, the number density of intrinsic charge carriers at 27^0C is $1.5\times10^{16}/m^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5\times10^{22}/m^3$. The electron density in the doped semiconductor is $_____\times10^9/m^3$.

1) 5 **2)** 15 **3)** 45 **4)** 1

35. In a coil, the current changes from -2A to +2A in 0.2 sec and induces an emf of 0.1 v. The self inductance of the coil is

1) 5 mH **2)** 1mH **3)** 2.5mH **4)** 4mH

36. Three blocks A,B and C are pulled on a horizontal smooth surface by a force of $80N$ as shown in figure



The tensions T_1 and T_2 in the strings are respectively

1) 40N, 64N **2)** 60N,80N **3)** 88N, 96 N **4)** 80 N, 100N

37. The force required to stretch a wire of cross section $1cm^2$ to double its length will be : (Given young's modulus of the wire= $2\times10^{11}\,N/m^2$)

1) $1\times10^7\,N$ **2)** $1.5\times10^7\,N$ **3)** $2\times10^7\,N$ **4)** $2.5\times10^7\,N$

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38. Match List I with list II

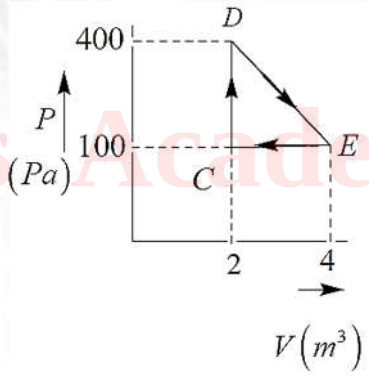
List –I	List –II
A) $\int \overline{B}.\overline{dl} = \mu_0 i_c + \mu_0 \in_0 \frac{d\phi_E}{dt}$	I) Gauss's law of electricity
B) $\int \overline{E}.\overline{dl} = \frac{d\phi_B}{dt}$	II) Gauss's law of magnetism
C) $\int \overline{E}.\overline{dA} = \frac{Q}{\in_0}$	III) Faraday's law
D) $\int \overline{B}.\overline{dA} = O$	IV) Ampere- Maxwell's law

Choose the correct answer from the options given below

1) A-II,B-III,C-I,D-IV **2)** A-IV,B-I,C-III,D-II

3) A-IV,B-III,C-I,D-II **4)** A-I,B-II,C-III,D-IV

39. A thermodynamic system is taken through cyclic process. The total work done in the process is



1) 100 J **2)** 300 J **3)** zero **4)** 200 J

40. A pulse of light of duration 100ns absorbed completely by a small object initially at rest. Power of the pulse is 30 mW and the speed of light is $3\times10^8 ms^{-1}$.The final momentum of the object is.

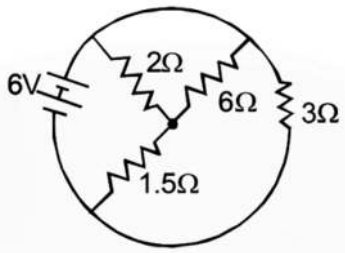
1) $0.3\times10^{-17} Kg\,ms^{-1}$ **2)** $3.0\times10^{-17} Kg\,ms^{-1}$

3) $1.0\times10^{-17} Kg\,ms^{-1}$ **4)** $9.0\times10^{-17} Kg\,ms^{-1}$

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41. The total current supplied to circuit by the battery is



1) 4A **2)** 2A **3)** 1A **4)** 6A

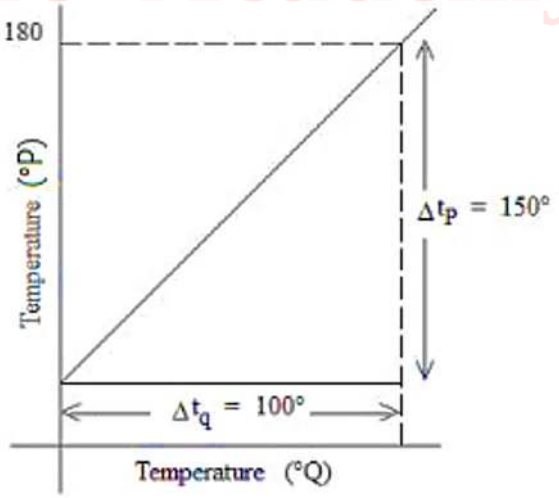
42. The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of rms velocities will be

1) $\frac{25}{4}$ **2)** $\frac{2}{5}$ **3)** $\frac{5}{2}$ **4)** $\frac{4}{25}$

43. The ratio of maximum acceleration to maximum velocity in a simple harmonic motion is $10S^{-1}$. At $t=0$ the displacement is 5m. What is the maximum acceleration? The initial phase is $\frac{\pi}{4}$

1) $500 m/s^2$ **2)** $500\sqrt{2}\, m/s^2$ **3)** $750\ m/s^2$ **4)** $750\sqrt{2}\ m/s^2$

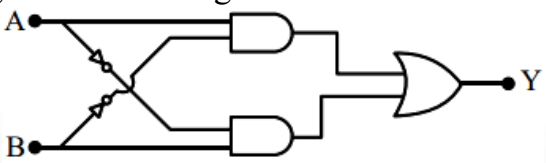
44. The graph between two temperature scales P and Q are shown in the graph. Between upper fixed point and lower fixed point there are 150 equal divisions of scale P and 100 divisions on scale Q. The relationship for conversion between the two scales is given by



1) $\frac{t_q}{150}=\frac{t_p-180}{100}$ **2)** $\frac{t_q}{100}=\frac{t_p-30}{150}$ **3)** $\frac{t_p}{180}=\frac{t_p-40}{100}$ **4)** $\frac{t_p}{100}=\frac{t_q-180}{150}$

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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTH RANK 1
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45. The truth table of the given circuit diagram is



1)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

3)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. An electron moves through an uniform magnetic field $\overline{B}=\left(B_0\hat{i}+2B_0\hat{j}\right)T$. At a particular instant of time the velocity of electron is $\overline{u}=\left(3\hat{i}+5\hat{j}\right)m/s$. If the magnetic force acting on electron is $\overline{F}=5e\hat{k}\,N$, where e is the charge of electron, then the value of B_0 is _____T.

47. When a dc voltage of 100 V is applied to an inductor , a dc current of 5A flows through it, when an ac voltage of 200V peak value is connected to inductor, its inductive reactance is found to be $20\sqrt{3}\Omega$. The power dissipated in the circuit is _____W.

48. A body of mass 2kg is driven by an engine delivering a constant power of 1J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m)_____

49. In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius $0.5A^0$. If the speed of electron is $2.2\times10^6\,m/s$,then the current associated with the electron will be _____$\times10^{-2}$ mA

[Take $\pi=\frac{22}{7}$]

50. The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2 .If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$ then the value of x will be _____

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. If uncertainty in momentum and position for A moving particle of mass 'm' is same than calculate uncertainty in it's velocity (h= plank's constant)

1) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ **2)** $\frac{1}{2\pi}\sqrt{\frac{h}{m}}$ **3)** $\frac{1}{m}\sqrt{\frac{h}{2\pi}}$ **4)** $\frac{1}{\pi}\sqrt{\frac{h}{2m}}$

52. Statement:-I: A reaction which is spontaneous and accompanied by decrease of randomness must be exothermic

Statement-II:- All exothermic reactions are accompanied by decrease of randomness

1) Statement – I is true, statement –II is true, statement II is a correct explanation for statement –I

2) Statement – I is true, statement –II is true, statement II is not a correct explanation for statement –I

3) Statement- I is true statement-II is false

4) Statement- I is false statement-II is true

53. Statement –I: For a reaction at equilibrium, the Gibb's free energy of reaction is minimum at constant temp, and pressure

Statement-II:- The Gibb's free energy of both reactants and products increase and become equal at equilibrium

1) Statement – I is true, statement –II is true, statement II is a correct explanation for statement –I

2) Statement – I is true, statement –II is true, statement II is not a correct explanation for statement –I

3) Statement- I is true statement-II is false

4) Statement- I is false statement-II is true

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54. Assertion: Molarity of pure water is 55.55M at 298 K

Reason:- Molarity is temperature dependent

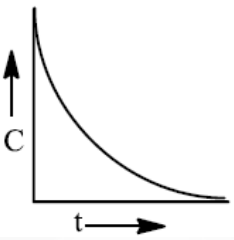
1) Both (A) and (R) are true but (R) is the correct explanation of (A)

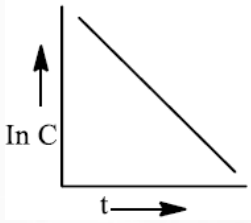
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

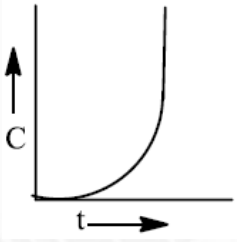
3) (A) is true but (R) is false

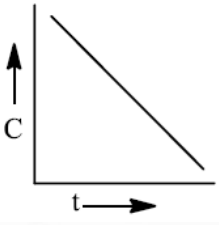
4) A) is false but (R) is true

55. The plot between concentration verses time for a zero order reaction is represented by

1) 

2) 

3) 

4) 

56. The $E^0 At\ 25^0 C$ for the following reaction is 0.22 V calculate the equilibrium constant

$At\ \ 25^0 C\ \ H_{2(g)} + 2AgCl_{(s)} \rightarrow 2Ag_{(s)} + 2HCl_{(Aq)}$

1) 2.8×10^7 **2)** 5.2×10^8 **3)** 5.2×10^6 **4)** 5.2×10^3

57. Assertion:- An increase in surface area increases the rate of evaporation

Reason:- Stronger the intermolecular attraction forces, faster is the rate of evaporation at a given temperature,

1) Both (A) and (R) are true but (R) is the correct explanation of (A)

2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

3) (A) is true but (R) is false

4) A) is false but (R) is true

58. The reason for the stability of Gd^{+3} ion is

1) $4f$ subshell-half filled

2) $4f$ subshell-completely filled

3) passes the general electronic configuration

4) $4f$ subshell-empty

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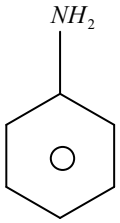
59. The effective Atomic Number of $Cr(Z=24)$ in $\left[Cr(NH_3)_6\right]Cl_3$ is

1) 35 **2)** 27 **3)** 33 **4)** 36

60. Both geometrical and optical isomerism are shown by

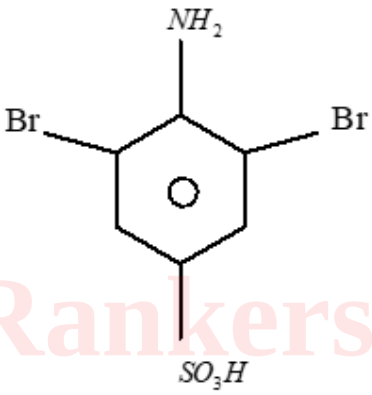
1) $\left[Co(en)_2Cl_2\right]^+$ **2)** $\left[Co(NH_3)_5Cl\right]^{+2}$ **3)** $\left[Co(NH_3)_4Cl\right]^{+1}$ **4)** $\left[Cr(Ox)_3\right]^{3-}$

61.

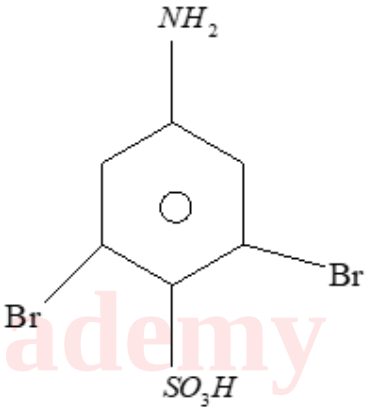


$\xrightarrow[453-473\,K]{conc\,H_2SO_4} \times \xrightarrow[Excess]{Br_2/H_2O} Y$ the product 'Y' is

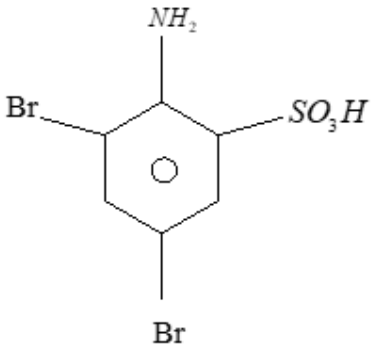
1)



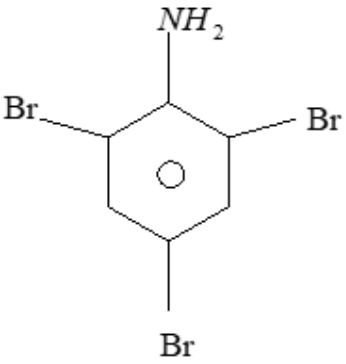
2)



3)

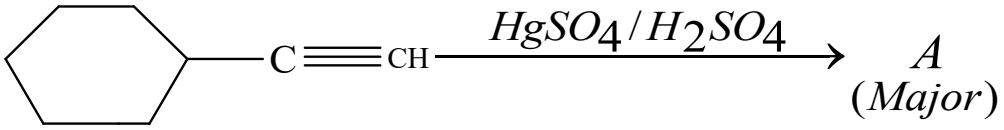


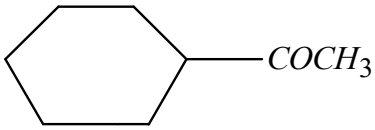
4)

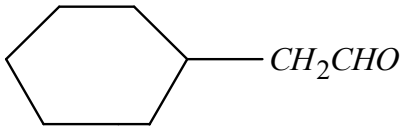


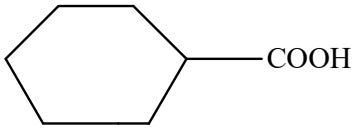
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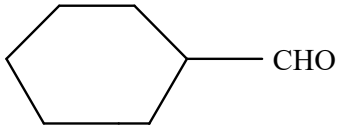
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62.  'A' is

1) 

2) 

3) 

4) 

63. Magnetic moments of $V(Z=23), Cr(Z=24)$ $Mn(Z=25)$ are x, y and z respectively hence

1) $x=y=z$ **2)** $x<y<z$ **3)** $x<z<y$ **4)** $z<y<x$

64. In PO_4^{3-} P-O Bond order is

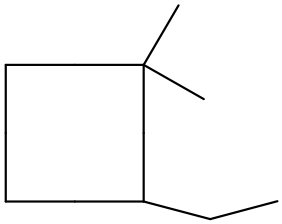
1) 1.25 **2)** 2 **3)**-0.75 **4)** -3

65. Glucose and galactose differ in configuration at

1) C-1 **2)** C-2 **3)** C-3 **4)** C-4

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66. IUPAC name of the following compound is



1) 2- Ethyl 1,1 di methyl cyclobutane **2)** 1- Ethyl 2,2 dimethyl cyclobutane

3) 1- Ethyl 4,4 dimethyl cyclobutane **4)** 4,4 dimethyl 1- ethyl - cyclobutane

67. When A hydro carbon 'A' undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water what is the molecular formula of A?

1) C_9H_8 **2)** $C_{11}H_4$ **3)** C_5H_8 **4)** $C_{11}H_8$

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68. Match the column:

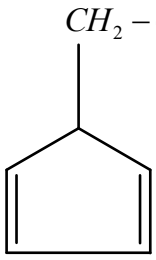
	Column-I E_2 **reactions** **Elimination biomolecular**		Column- II **Number of products** **(Including stereoisomerism)**
a)	$CH_3-CH_2-CH_2-CH_2-Br \xrightarrow{alc.KOH}$	p)	1
b)	$CH_3-\underset{\displaystyle Br}{\underset{\vert}{CH}}-CH_2-CH_3 \xrightarrow{alc.KOH}$	q)	2
c)	$CH_3-\underset{\displaystyle Br}{\underset{\vert}{\overset{\displaystyle CH_3}{\overset{\vert}{C}}}}-CH_2-CH_3 \xrightarrow{alc.KOH}$	r)	3
d)	$Ph-CH_2-\underset{\displaystyle Br}{\underset{\vert}{CH}}-CH_2-CH_3 \xrightarrow{alc.KOH}$	s)	4

1) $a-p,b-r,c-q,d-s$ **2)** $a-s,b-r,c-q,d-p$

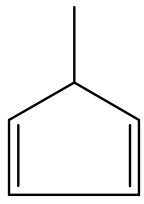
3) $a-r,b-p,c-q,d-s$ **4)** $a-p,b-q,c-r,d-s$

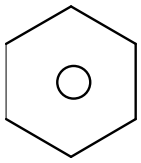
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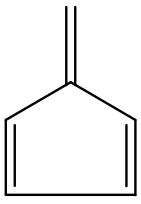
69.

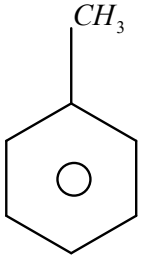
Product (A) is 

$\xrightarrow[\Delta]{H^{(+)}}$ A

1) 

2) 

3) 

4) 

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70. Nessler's reagent is uses to detect

1) CrO_4^{-2} **2)** PO_4^{3-} **3)** MnO_4^- **4)** NH_4^+

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The radius of an orbit of H-Atom is 0.2116 nm, kinetic energy (in ev) of electron present in this orbit is _____

72. If 4 Amp current produced in fuel cell use for the deposition of Ag^+ in 1 Lit $AgNO_3(Aq)$ solution for 241.25 sec using 'pt' electrode calculate the P^{OH} solution after electrolysis at 25^0C

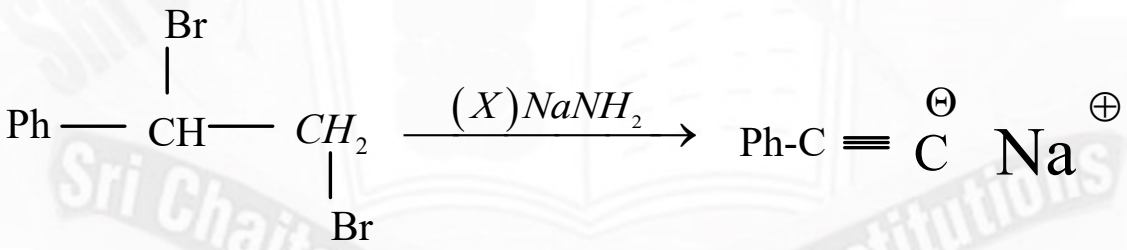
73. How many molecules among the following have zero dipole moment?

NH_3, BF_3, NF_3, CCl_4

74. Among the following oxides how many of them are suboxides?

a) C_3O_2 b) N_2O c) NO_2 d) CO e) Fe_3O_4 f) KO_2

75.



Find out the value of (X)

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